Large Model Course Project Description Document

Project 1: SUSTech Campus Knowledge Base Construction with RAG Technology

1. Project Background and Objectives

1.1 Project Background

Retrieval-Augmented Generation (RAG) is one of the most mainstream and low-cost technical solutions for large model implementation. Without modifying model parameters, it only needs to connect to an exclusive knowledge base, which can alleviate problems such as "outdated knowledge, inaccurate answers, and lack of domain knowledge" of large models at a low cost.

Campus-related information is usually scattered across different sources such as official websites, announcements, department introductions, living guides, admission brochures, and FAQs, resulting in low efficiency for students and faculty to query. By building a campus-specific RAG knowledge base, accurate and fast Q&A for campus-related questions can be achieved.

1.2 Core Objectives

- Understand the core principles and complete workflow of RAG;
- Distinguish the differences between RAG and model fine-tuning, and clarify the applicable scenarios of RAG;
- Master the methods of collecting, cleaning, slicing, and vectorizing campus text data;
- Complete the full link of "user question → knowledge retrieval → context splicing → model generation";
- Build an interactive campus knowledge base Q&A system;
- Try to optimize the system effect through retrieval parameters, slicing methods, re-ranking, and other methods.

2. Resources Required for the Project

- Dataset: Campus Knowledge Base

  Can be collected from public sources such as SUSTech official website, campus brochures, department websites, living guides, admission materials, etc.
- Test Set: Design more than 10 test questions based on the knowledge base for effect comparison between non-RAG and RAG solutions
- Recommended Model: Qwen3.5 or other lightweight open-source models

3. Task Requirements

Task	Specific Requirements
Data Vector Database Construction	Obtain SUSTech's public campus information, complete text cleaning, slicing, vectorization, and database construction
Local Deployment of Large Model	Deploy a lightweight large model locally, implement model inference, and test that the basic dialogue function is normal
RAG and Large Model Integration	Connect the full process of "user question → knowledge base retrieval → context splicing → large model answer generation"
System Effect Comparison	Compare at least the difference in answer effects between "direct Q&A without RAG" and "after adding RAG"It is recommended to display at least the following content: Accuracy before and after adding the RAG solution. Comparison of answer examples before and after RAG implementation. Error analysis
Presentation and Project Report	Introduce the problem background, experimental process, knowledge base construction method, RAG connection scheme, problems encountered, solutions, effect comparison, and summary

4. Assessment Methods

- Assessment Content 1: Project Report
- Assessment Content 2: Presentation and On-site Demonstration

  During the presentation, it is required to demonstrate the Q&A system on-site and answer campus-related questions

---

Project 2: Teaching Assistant Fine-tuning Based on Synthetic QA Data

1. Project Background

In the course teaching scenario, large models can undertake tasks such as course Q&A and concept explanation, but when using general models directly, there are often problems such as unstable coverage of course knowledge, inconsistent answer styles, and insufficient sensitivity to specific course content.

Focusing on the scenario of "course teaching assistant", this project requires students to first use more powerful open-source models to generate course-related QA pairs, then clean and filter the data, and finally perform lightweight fine-tuning on small models using the cleaned data, and compare the Q&A effects before and after fine-tuning.

The core process of the project is:

Course Materials Collation → QA Data Generation → Data Cleaning → SFT Fine-tuning → Effect Comparison Before and After Fine-tuning

2. Project Objectives

After completing this project, students should be able to:

1. Learn to construct teaching Q&A data based on course materials;
2. Learn to generate QA pairs using existing open-source models or APIs;
3. Learn to clean and filter synthetic data;
4. Learn to perform lightweight fine-tuning on small models using LoRA / QLoRA;
5. Compare the performance of the model on course Q&A tasks before and after fine-tuning.

3. Project Tasks

Task	Specific Requirements
Course Material Collation	Select 1 university course or 1 course module, and collate lecture notes, courseware text, experiment guidelines, or FAQs
QA Data Generation	Use a more powerful model to generate course-related QA pairs, and the questions should cover the core knowledge points of the course
Data Cleaning	Deduplicate, unify the format, eliminate error samples, and filter out low-quality samples from the generated data
Model Fine-tuning	Perform lightweight fine-tuning on small models using cleaned QA data, LoRA / QLoRA is recommended
Effect Comparison	Show the difference in the performance of the model on course Q&A tasks before and after fine-tuning
Report and Presentation	Explain the data source, generation method, cleaning strategy, model settings, and experimental results

4. Data Requirements

Each group needs to collate basic materials around one course or one course module, and optional sources include:

- Course lecture notes
- Exported text from courseware
- Experiment guidelines
- Course FAQs
- Textbook chapter summaries

It is recommended to select course materials with **clear boundaries of knowledge points**.

Recommended Question Types

- Definition-based questions
- Fact-based questions
- Comparison-based questions
- Condition-based questions
- Simple application-based questions

Recommended Data Scale

- Original synthetic QA: 500 ~ 2000 pieces
- Cleaned training data: 300 ~ 1000 pieces
- Test set: 30 ~ 100 pieces

5. Model and Fine-tuning Requirements

Recommended Models

- Qwen2.5-0.5B / 1.5B / 3B
- Qwen3.5-0.8B / 2B

Recommended Fine-tuning Methods

- LoRA
- QLoRA

Minimum Experimental Closed Loop

Each group needs to complete at least:

- Baseline before fine-tuning
- Model after fine-tuning
- A set of effect comparisons

6. Evaluation Requirements

It is recommended to display at least the following content:

- Accuracy on the test set before and after fine-tuning
- Comparison of answer examples before and after fine-tuning
- Simple error analysis

Analysis can be conducted from the following perspectives:

- Whether the answer is correct
- Whether the answer is consistent with the course materials
- Which questions have significant improvement
- Which questions still perform poorly

7. Report Requirements

The project report should include at least:

1. Experimental background and objectives
2. Dataset introduction
    - Data source
    - Data scale
    - Cleaning steps
3. Model and fine-tuning strategy
    - Model selection
    - Data generation method
    - Fine-tuning settings
4. Experimental results
    - Effect comparison before and after fine-tuning
5. Result analysis
6. Experimental summary

8. Reference Materials

Data Generation

- Self-Instruct：https://arxiv.org/abs/2212.10560
- Self-QA：https://arxiv.org/abs/2305.11952

Fine-tuning

- Hugging Face PEFT：https://huggingface.co/docs/peft/index
- LoRA：https://huggingface.co/docs/peft/main/en/conceptual_guides/lora

Model Download

- ModelScope：https://modelscope.cn/

Model API Platform

- Alibaba Cloud Tongyi Wanlian：https://api.aliyun.com/document/bailian

9. Final Submission Content

Each group needs to submit:

1. Project report
2. Presentation materials

Project 3: Multimodal Fine-tuning for Autonomous Driving Sequence Understanding

1. Project Background and Objectives

1.1 Project Background

Video sequences in autonomous driving or intelligent driving scenarios contain rich spatiotemporal information, such as:

- Whether the vehicle turns left, turns right, or goes straight;
- Whether it is in a state of following the car, changing lanes, decelerating, etc.;
- Whether surrounding traffic participants have an impact on the current driving behavior.

Judgment based solely on text or single-frame images is usually insufficient, while video sequences combined with text descriptions can provide more complete contextual information. This project expects students to complete a multimodal fine-tuning experiment of "input image + auxiliary text description, output current behavior analysis" based on the given dataset.

The focus of this project is not to pursue high-difficulty autonomous driving decision modeling, but to complete a **medium-difficulty image-text understanding and fine-tuning experiment**, and compare the effect differences of different open-source models after fine-tuning.

1.2 Core Objectives

- Understand the basic usage of multimodal large models in video understanding tasks;
- Learn to organize video clips and text descriptions into data formats suitable for model training;
- Complete lightweight fine-tuning of at least one open-source multimodal model;
- Output behavior analysis results of the current sequence, such as left turn, right turn, straight, lane change, etc.;
- Compare the result differences under different models or different fine-tuning settings;
- Analyze common error sources in video understanding tasks.

2. Resources Required for the Project

- Dataset: Dataset provided by the teacher
- Recommended Model Directions:
    - Qwen3.5 series
    - VideoLLaMA2 series
    - Other open-source models that can perform video / image-text understanding

3. Task Requirements

Task	Specific Requirements
Data Preprocessing	Complete video sampling, frame extraction, annotation format collation, and adapt to the input format of the selected model
Model Fine-tuning	Select at least one open-source multimodal model for fine-tuning; it is encouraged to compare two models or two settings
Result Comparison	Compare results before and after fine-tuning, or compare the performance of different models / different fine-tuning configurations
Presentation and Project Report	Explain the dataset processing method, model selection, training settings, effect comparison, problem analysis, and summary

4. Recommended Experimental Settings

It is recommended to compare at least one of the following two types of results:

Scheme A: Comparison of the same model before and after fine-tuning

- Baseline before fine-tuning
- Results after fine-tuning

Scheme B: Comparison of different models

- Results after fine-tuning of Model 1
- Results after fine-tuning of Model 2

5. Assessment Methods

- Assessment Content 1: Project Report
- Assessment Content 2: Presentation

  During the presentation, it is recommended to display:

  - Input video clips
  - Corresponding text descriptions
  - Model output behavior analysis
  - Result comparison of different models or before/after fine-tuning

Appendix: Unified Submission Requirements (Applicable to the above three projects)

1. Project Report

Must include the following basic structure:

1. Project Background and Objectives
2. Dataset / Data Source Introduction
3. Model and Method Design
4. Experimental Process
5. Result Display
6. Problem Analysis
7. Summary

2. Presentation Requirements

Each group needs to complete:

- Presentation / PPT
- On-site presentation or case demonstration

3. Unified Recommendations

- Prioritize ensuring the project can run through
- Prioritize ensuring the existence of a baseline
- Prioritize ensuring results are reproducible
- It is not recommended to blindly stack complex modules

Project	**Minimum Hardware Requirements**
Project 1	CPU，16GB RAM
Project 2	12GB GPU，16GB RAM
Project 3	12GB GPU，16GB RAM